

Fourier transform activity

Student activity

Name _____ Date _____

- Construct graphs of data using the ***FourierTransform_ActivityGraph.ods*** file.
- Each graph should have data that appears random but has data points at one or more regular time intervals.
- Time axis from 0 s to 60 s
- Each graph contains two signals at the periods indicated below, with noise included in the signal. These are created in the *FourierTransform_ActivityGraph.ods* file.

1. (3, 7)

2. (4, 7)

3. (3, 8)

4. (4, 9)

5. (4, 6)

6. (3, 5)

7. (5, 9)

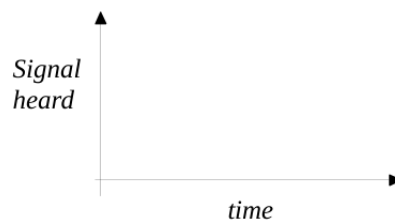
8. (5, 7)

9. (6, 9)

10. (5, 8)

50 s on time-axis = 7.50" $\rightarrow$ 1 s = 0.15"

- Intro
- Hand out a blank graph of signal vs. time to each student.



- Demonstrate GNU Radio sound of single, clean periodic pulses at 1.0 s intervals. Ask students to sketch what they think the pulse would look like on the graph.
- Have students discuss their graph with another student; then discuss as class:
- Have a volunteer sketch their graph on the board
- Why does your graph represent what you heard?
- How is the *period* of a wave represented in your graph?
- This is an example of a periodic signal.

• Signals that are not noise usually have some sort of pattern that is repetitious - periodic.

• Review the *period*, *amplitude*, and *frequency* of this signal, and how each is represented in a signal.

- Then demonstrate GNU Radio pulses with noise added. Ask students to sketch what they think the signal would look like in this case.
- Students compare with a neighbor.
- Then discuss as class:

• What's different?

• What's the same?

• Noise is usually not periodic. Recall from yesterday: Signals carry information.

When processing signals, it is necessary to distinguish the signal from the noise.

• Recall Tad's presentation with the photo filtering. This is filtering of amplitude.

• Another method scientists use is filtering by frequency. Scientists have methods to filter signals by frequency (or period).

- Signal Processing Activity
- Demonstrate process with the teacher graph

• explain masks.

• demonstrate sliding process to identify periods.

• explain that a periodic signal must fill at each period translation.

• explain how noise shows up – signals less than 1.0 (usually), and not periodic.

• Explain offset.

- Group students into pairs.
- Handout signal sheets and masks. Explain that they are to determine the periods in each signal graph; then replot the data showing only those periodic signals, without noise.
- Have students sketch the Signal vs. Frequency graph that corresponds to their
- Post Activity Discussion
- What challenges are there determining the periodic signals in the total signal?
- For the 3 s and 4 s signals, harmonics show up. How do you think we can distinguish whether the actual signal is the 3 s, 6 s, or 9 s?
- Many signals are a mixing of multiple frequencies – next activity with Howard.
- Scavenger hunt clue.
- Motivation: Explain/remind students that waves are repetitious, e.g. water waves oscillate periodically. Information can be carried in the pattern, e.g. the morse code activity done previously.
- Show sample graphs:
- Example showing sum of sinusoidal signals. What periodic waves are in this?
- Example with periodic pulse signal in noise.

- What repetitious (periodic) waves in this signal?
- Demonstrate masking method for determining the periodic waves in this.
 - Lay the mask over the graph and slide the mask until each opening aligns with the data at one second intervals.
 - By sliding the mask to match up with the data at each one second interval the period of the signal can be determined.
 - Pair students up. Hand out graphs and masks to each group.
 - Students are to use the mask method to determine the period of any waves in their graph.
 - Have each group sketch what the graph would look like if there were no noise.
 - Extension: On a signal vs. frequency graph, have each group plot signal vs. frequency.
 - Scavenger hunt:
 - column (0 – 9) determined by amplitude of stronger signal
 - row (X – R) determined by frequency of weaker signal; use a table to convert frequency to a letter.